

- 3 Type 1 diabetes is a disease that occurs when the pancreas stops producing insulin. Type 1 diabetes occurs most commonly in children and is often the result of the immune system destroying beta cells in the pancreas. Beta cells are the cells in the pancreas that make insulin.

*Rotavirus* is a genus of RNA viruses that can cause infections in people throughout most of the world. Infection by *Rotavirus* causes sickness and diarrhoea in babies and young children, and may damage the pancreas. *Rotavirus* infections are also associated with the development of type 1 diabetes in children.

In May 2007, a vaccination against *Rotavirus* was introduced in Australia. All babies aged from 2 months to 4 months were offered the *Rotavirus* vaccine.

Some scientists decided to investigate whether vaccination against *Rotavirus* had reduced the number of children with type 1 diabetes in Australia.

- (a) Identify the **independent** variable **and** the **dependent** variable in this investigation.

independent variable .....

dependent variable .....

[2]

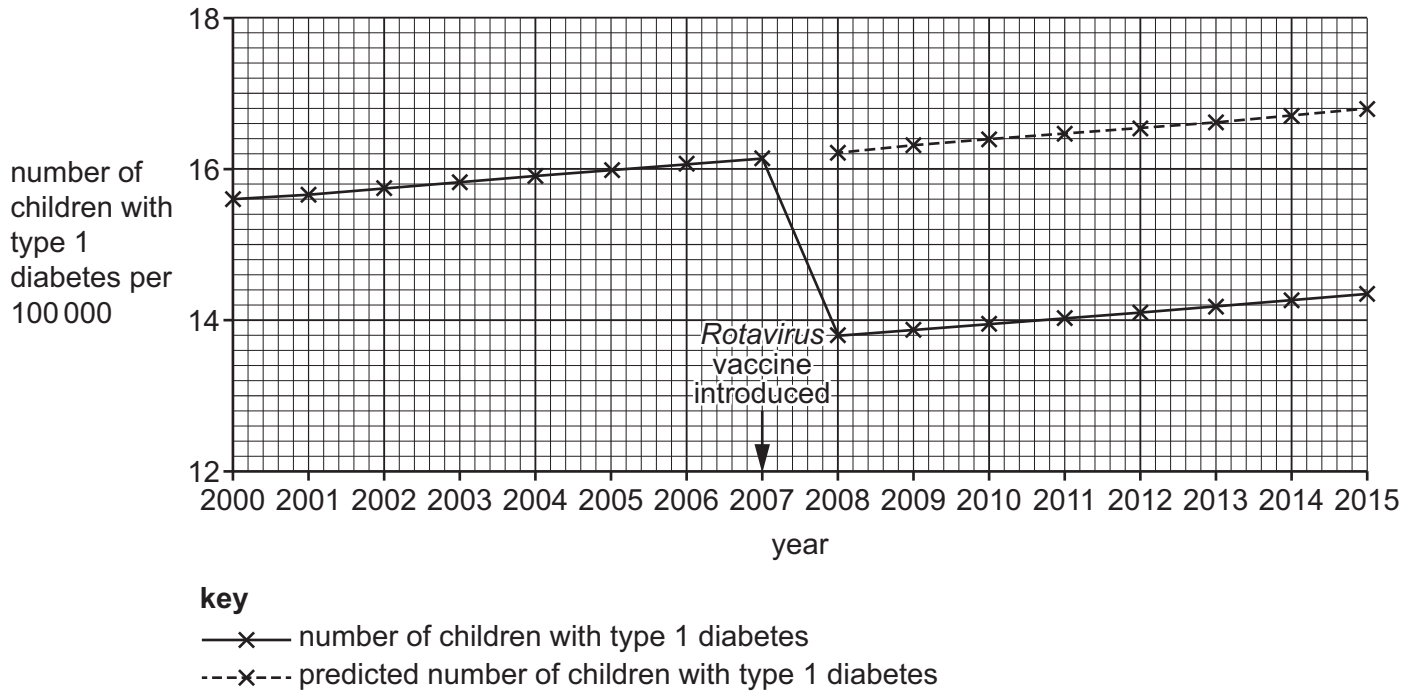
- (b) The scientists used data that had been collected from 2000 to 2015. For each year from 2000 to 2015, the scientists used the data to:

- find the number of children aged from 0 years to 4 years
- find the number of children aged from 0 years to 4 years with type 1 diabetes
- calculate the number of children aged from 0 years to 4 years with type 1 diabetes per 100 000 children that were aged from 0 to 4 years.

The scientists repeated this analysis for children aged from 10 years to 14 years.

- (i) Fig. 3.1 shows the number of children aged from 0 years to 4 years with type 1 diabetes per 100 000 for each year from 2000 to 2015.

The predicted number of children aged from 0 years to 4 years with type 1 diabetes per 100 000, if the *Rotavirus* vaccine had **not** been introduced, is also shown from 2008 to 2015.



**Fig. 3.1**

State **two** conclusions that can be made from the data shown in Fig. 3.1.

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..... [2]

- (ii) Fig. 3.2 shows the mean number of children with type 1 diabetes in Australia, in each of the two different age groups analysed, before and after the introduction of the *Rotavirus* vaccine in 2007. Error bars show 95% confidence intervals (95% CI).

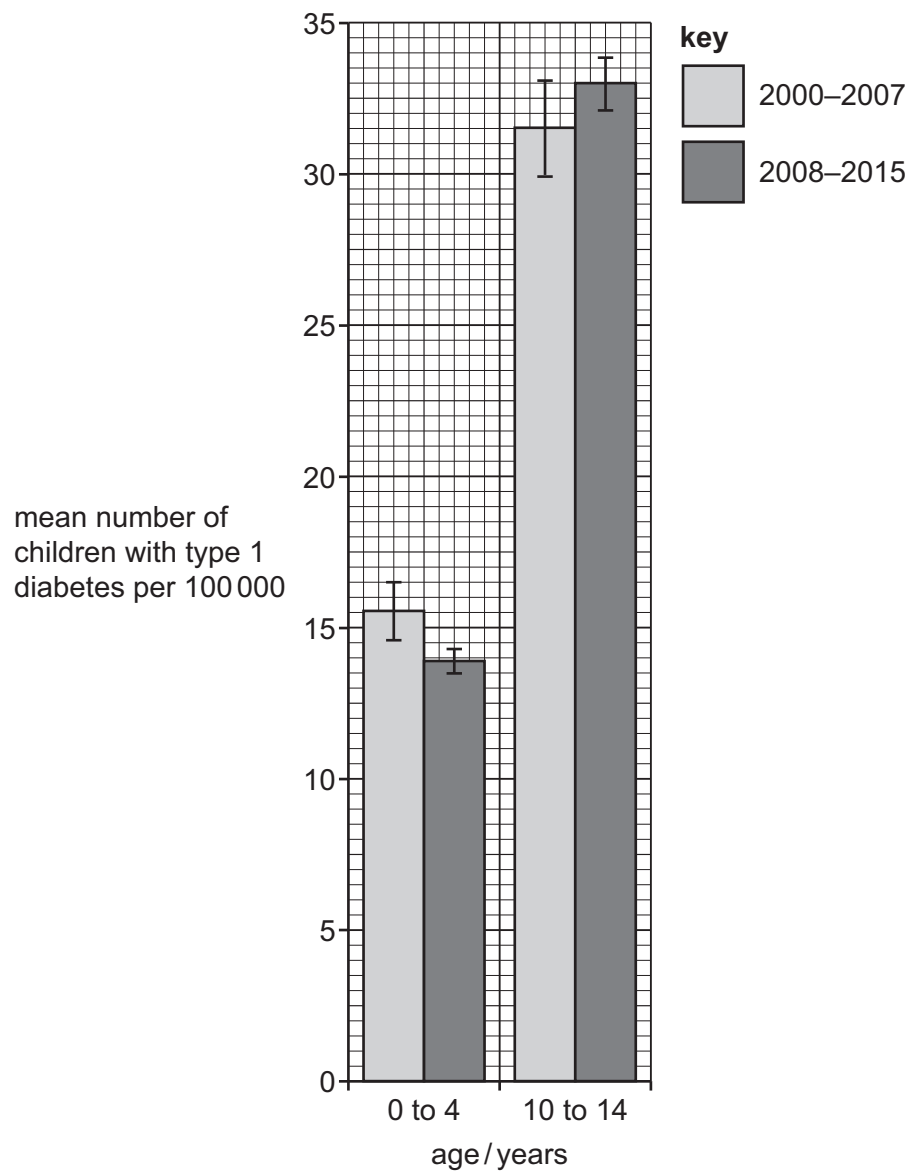


Fig. 3.2



Some students concluded that vaccination against *Rotavirus* decreased the number of children with type 1 diabetes per 100 000.

Discuss whether the information shown in Fig. 3.2 supports this conclusion.

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[Total: 7]